

Smart Poultry Health Monitoring and Disease Detection System Using IoT Sensors

This project aims to design a Smart Poultry Health Monitoring and Disease Detection System that automatically monitors and controls various parameters essential for the health and productivity of hens. The system leverages IoT sensors to detect environmental changes, disease symptoms, and automate feeding and cleaning tasks. It reduces human labor, prevents disease outbreaks, and ensures optimal environmental and feeding conditions.

Main Functionalities

1. Measure and control the temperature and humidity inside the coop using DHT11 or LM35 sensors. 2. Automatically regulate temperature using a relay-controlled fan or heating bulb. 3. Measure the quantity of food using a load cell (HX711 module). 4. Monitor water level and automatically pump water using a water level sensor and mini pump. 5. Dispense medicine (mixed with water or feed) using a servo motor-controlled dispenser. 6. Detect diseases by analyzing irregular temperature changes, reduced movement, or low feed/water intake. 7. Clean the coop periodically using a mini cleaning motor or fan. 8. Monitor air quality using an MQ-135 gas sensor to prevent respiratory diseases. 9. Track hen activity using an ultrasonic or IR sensor to detect inactivity or stress. 10. Send alerts to a mobile app or display on an LCD when abnormal conditions are detected.

Sensors and Components Used

Component	Function
ESP32 / Arduino Uno	Main microcontroller controlling all operations
DHT11 / DHT22 / LM35	Measures temperature and humidity
Load Cell + HX711	Measures feed quantity
Water Level Sensor	Monitors water availability
Mini Water Pump	Supplies water automatically when low
Servo Motor	Dispenses food or medicine automatically
Relay Module	Controls fan, bulb, or pump based on conditions
MQ-135 Gas Sensor	Detects ammonia and air quality in the coop
Ultrasonic Sensor	Detects motion or presence of hens
IR Temperature Sensor	Measures body temperature of hens
16x2 LCD Display	Displays real-time system status and alerts
Buzzer / LED	Alerts when abnormal conditions are detected

How the System Works

The Smart Poultry Monitoring System continuously collects data from various sensors installed in the chicken coop. The ESP32/Arduino microcontroller processes these inputs and takes automatic actions such as switching on fans or heaters, pumping water, or activating dispensers. If irregular patterns are detected—like reduced activity, high temperature, or poor air quality—the system alerts the farmer through the LCD display or mobile app. These signs may indicate disease or discomfort. By integrating temperature, motion, feeding, and air quality sensors, the system effectively predicts and prevents common poultry diseases, ensuring improved health and productivity.